Predictive learning explains emergence of objects in infants

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Abstract

How do young animals learn to parse visual scenes into objects with so little experience? Human infants initially rely on motion cues to segregate objects, while sensitivity to static shape cues emerges only later — mirroring the earlier maturation of the dorsal visual stream relative to the ventral stream. Here we introduce a two-stream neural network in which a shape pathway and a motion pathway cooperate to predict future visual input. Trained on videos of moving objects, the model first discovers objects through coherent motion, then develops sensitivity to static shape cues, recapitulating the developmental trajectory observed in infants. These results demonstrate that cross-stream predictive learning is sufficient to produce object representations from a realistic amount of visual experience, without requiring innate perceptual priors.

Introduction

Animals parse visual scenes into objects with remarkably little experience — human infants can group moving surfaces into coherent objects by four months of age. Yet how this capacity develops from raw sensory input remains an open question.

Across species, motion cues precede static shape cues in development. Four-month-old infants group surfaces that move together into a single object but fail to use static alignment cues such as good continuity (Kellman & Spelke, 1983), and non-motion synchronous changes do not produce the same grouping (Jusczyk et al., 1999). Adults whose congenital blindness was surgically corrected show the same pattern: object parsing initially required motion, and sensitivity to static cues emerged only after months of visual experience (Ostrovsky et al., 2009). This developmental order mirrors the earlier maturation of the dorsal visual stream relative to the ventral stream (Ciesielski et al., 2019), suggesting that motion processing scaffolds the later acquisition of static grouping cues.

Why should motion be privileged? The brain is thought to continuously predict its future sensory input (Caucheteux et al., 2023; Clark, 2013; Rao & Ballard, 1999), and predicting the next visual frame requires estimating what moves and where. We hypothesize that the ventral and dorsal streams cooperate to make this prediction: the dorsal stream estimates motion, while the ventral stream learns to segment the regions that move coherently — and in doing so, acquires static shape representations as a byproduct. Consistent with this, infant visual learning is impaired when motion is made temporally unpredictable (Kellman, 1984).

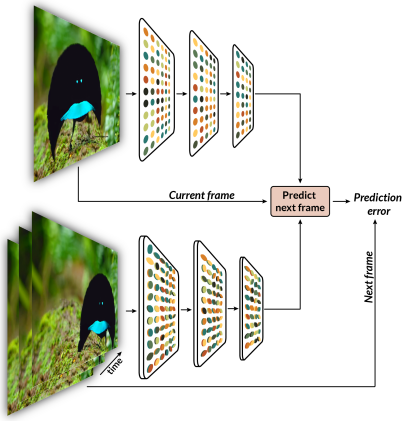

Figure 1: **Two stream model of vision.** The model has a separate form and motion pathway which work together to predict the next frame. The prediction error calculated using the predicted and ground truth frames are used to train the model.

To test this, we build a two-stream neural network whose ventral and dorsal pathways jointly predict the next video frame. Trained on synthetic videos of moving objects, the model first segments objects by coherent motion and later develops sensitivity to static shape cues, recapitulating the developmental trajectory observed in infants.

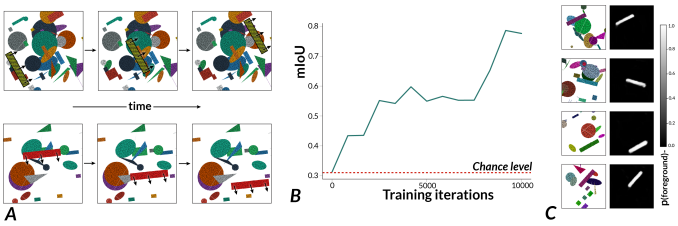

Figure 2: **Predictive learning begets common fate. A.** Example frames of videos used to train the model. **B.** Object segmentation accuracy over iterations of prediction error minimization. **C.** Example object segmentation predictions.

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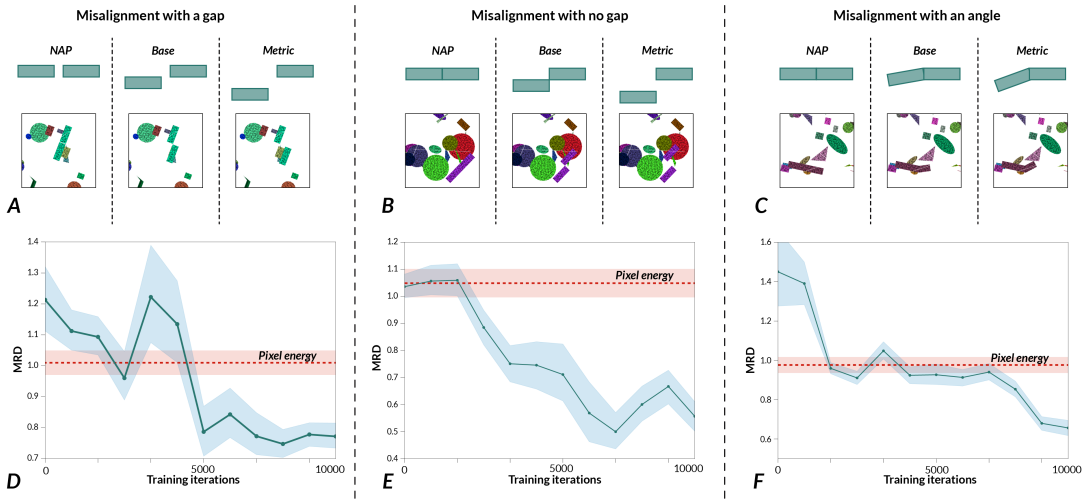

Figure 3: **Sensitivity to static grouping cues also emerge with prediction error minimization. A-C.** Example triplets generated to determine sensitivity to static cues in the model. **D-F.** Mean relative dissimilarity (MRD) is defined as the ratio of representational dissimilarities of Base & Metric pairs to NAP & Base pairs.

Results

A two stream model of vision

Motivated by the two-streams hypothesis (Goodale & Milner, 1992), our model has three components (Fig. 1). An early visual pathway $\phi_{\theta e}$, loosely corresponding to retina through V1, transforms consecutive frames x_t, x_{t+1} into feature maps y_t, y_{t+1} through a hierarchy of simple- and complex-cell-like operations. These features feed into two parallel streams. A shape stream $\phi_{\theta v}$, modeling ventral areas V2/V4/IT, processes a single frame through a hierarchy of feature-selective units and produces a spatial segmentation map via competitive selection: $f_v = \text{softmax}(\phi_{\theta v}(y_t))$. A motion stream $\phi_{\theta d}$, modeling dorsal areas MT/MST, applies direction-selective spatiotemporal filters to the pair (y_t, y_{t+1}) to estimate what has moved: $f_d = \phi_{\theta d}(y_t, y_{t+1})$. The shape and motion signals are combined ($h = f_v \circ f_d$) and used alongside the current frame to generate a prediction of the next frame $\hat{x}_{t+1}$. All parameters are learned jointly by minimizing L^2 prediction error between $\hat{x}_{t+1}$ and x_{t+1}.

Predictive learning begets common fate.

We train the model on 2500 synthetic videos (20 frames each) in which a textured rectangle moves over a background of random shapes (Fig. 2A). The only cue identifying the foreground as a single object is coherent motion; the only static cue is its rectangular shape. With 10,000 iterations of training (minimizing the prediction error), the shape stream's output f_v increasingly aligns with the ground-truth segmentation mask – a mean Intersection over Union (mIoU) score of around 0.8, well above chance-level (Fig. 2B–C), showing that the model discovers objects through motion without any segmentation supervision.

Predictive learning begets good continuity.

Human adults are more sensitive to changes in edge alignment (a non-accidental property, or NAP) than to equivalent pixel-level changes that preserve misalignment (Kim & Biederman, 2012). To test whether the model develops the same bias, we adopt the triplet paradigm of (Kim & Biederman, 2012): for each stimulus set, an aligned (NAP) shape, a misaligned (base) shape, and a more-misaligned (metric) shape are constructed so that pixel distances between adjacent pairs are equal (MRD ≈ 1 in pixel space). If the shape stream is sensitive to alignment, its representational distance for the NAP–base pair should exceed that for the base–metric pair, yielding MRD < 1. We find this across all three stimulus variants (Fig. 3D–F), indicating that sensitivity to static grouping cues emerges from predictive learning alone.

Conclusion

A two-stream model trained to predict the next visual frame first discovers objects through coherent motion and then develops sensitivity to static shape cues, recapitulating the developmental trajectory observed in human infants. These results show that the perceptual organization traditionally attributed to innate Gestalt priors can instead emerge from predictive learning over a realistic amount of visual experience. More broadly, they point to cross-stream prediction as a computational principle that may underlie the development of object perception in biological vision.

References

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